

Experiment no-02

2. Implement a basic finite state automaton that recognizes a specific language or pattern. In this example, we'll create a simple automaton to match strings ending with 'ab' using python.

PROGRAM:

```
def ends_with_ab(string):
    state = "q0"

    for ch in string:
        if state == "q0":
            if ch == 'a':
                state = "q1"
            elif ch == 'b':
                state = "q0"

        elif state == "q1":
            if ch == 'a':
                state = "q1"
            elif ch == 'b':
                state = "q2"

        elif state == "q2":
            if ch == 'a':
                state = "q1"
            elif ch == 'b':
                state = "q0"

    if state == "q2":
        return "Accepted (String ends with 'ab')"
    else:
        return "Rejected (String does not end with 'ab')"

text = input("Enter a string: ")
print(ends_with_ab(text))
```

OUTPUT:

```
... Enter a string: aaab
    Accepted (String ends with 'ab')
```